Available online at www.sciencedirect.com
ScienceDirect
Hormones and Behavior 50 (2006) 623–631
Hormones and Behavior
www.elsevier.com/locate/yhbeh

The impact of early adverse care on HPA axis development: Nonhuman primate models

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Received 12 May 2006; revised 15 June 2006; accepted 16 June 2006
Available online 15 August 2006

Abstract

This review presents supporting evidence that early disruptions in mother–infant relationship in primates, including infant maltreatment, are important risk factors for the development of psychopathology and pathophysiology during childhood and adolescence. Current research in this field is trying to identify important aspects of early adverse experiences such as the timing, frequency, duration, "perceived" intensity of the stressful or traumatic events, the role of social support (e.g., nurturing caregiver) in buffering the deleterious outcomes of early adversity, as well as the role of sex and genetic factors on individual variability in vulnerability. The use of nonhuman primate models of early adverse caregiving is helping to put the pieces of the puzzle together to fully understand the causes and consequences of similar experiences in humans. These models are essential to characterize the time course of biobehavioral alterations throughout development, using prospective, longitudinal studies performed under controlled experimental conditions and using invasive approaches that are unrealistic and unethical when studying human populations.
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Keywords: HPA axis; Infant maltreatment; Early adverse experiences; Monkey; Stress

Introduction

Nonhuman primates as animal models of early adverse caregiving

Nonhuman primates are the closest animal models of early adverse caregiving in humans. As in humans, caregiving is very involved and spans a considerable period of infant and juvenile development, with complex parent–offspring relationships. Like humans, monkeys and apes have a prolonged postnatal period of maturation and growth, when development of many neural systems can be influenced by experience. One important aspect of this early experience is the social environment, and in particular, the mother–infant relationship.

The rhesus monkey (*M. mulatta*) is the primate species most commonly used to study developmental effects of early social environment and, therefore, the main focus of this review. There are several reasons for this choice. One is their adaptive and ubiquitous nature that makes them survive and thrive in different environments, including captivity under laboratory conditions. But the most relevant reason is their highly social nature, with animals living in troops with moderately stable social organization, clear dominance matriarchal hierarchies, and close mother–infant relationships (Suomi, 2005; Hinde and Spencer-Booth, 1967).

The rhesus macaque infant spends the first month of life in close physical contact (mostly ventro-ventral) with its mother, who typically limits the infant's social interactions to immediate family members during this time. In fact, mother–infant pairs remain in physical contact for nearly a year, after which the next sibling is born and weaning materializes. After the end of the first year, the juvenile still maintains a close relationship with its mother and the mother continues to play an important role in providing protection and training the offspring to develop appropriate social behaviors. Therefore, for the infant rhesus monkey, the primary early social relationship is with its mother, and the integrity of this relationship is crucial for proper infant

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0018-506X/$ - see front matter © 2006 Elsevier Inc. All rights reserved.
doi:10.1016/j.yhbeh.2006.06.012

socioemotional, cognitive and neurobiological development (e.g., for reviews see Levine, 2005; Pryce et al., 2005b; Sanchez et al., 2001). An important aspect of this mother–offspring relationship in macaques is to enable the infant to control and modulate its physiological and behavioral responses to stress and its arousal levels by means of maintaining contact and proximity with the mother.

Thus, mother and infant rhesus monkeys form strong and enduring social bonds comparable to the one seen in humans (Bowlby, 1969; Hinde, 1974). Beginning around the second month of life infant macaques start exploring their environment and they increase the time spent engaged in social interactions (mostly play) as they grow up (Hinde and Spencer-Booth, 1967; Suomi, 2005). Although interactions with peers remain high during the juvenile period (1 to 3 or 4 years of age), the juvenile macaque maintains a tight social relationship with its mother, seeking physical contact with her under stressful situations and joining her (and soliciting her support) during social interactions, for example.

Many studies of the impact of mother–infant relationship disruption in nonhuman primates have used the permanent removal of the mother early in life ("maternal deprivation" models) or even from birth ("maternal privation") (Harlow et al., 1965; Harlow et al., 1971; for review see Sanchez et al., 2001). These models are more comparable to the severe early deprivation experienced by children reared in some orphanages (see Tarullo and Gunnar, this volume), and have provided invaluable information in highlighting the devastating consequences of maternal deprivation on many aspects of development. Despite their relevance, they are not the focus of this review because they do not adequately model aspects of early adverse caregiving such as those present in childhood maltreatment.

The role of the HPA axis in the response to stress

During threatening situations, the hypothalamic–pituitary–adrenal (HPA) axis and autonomic nervous system (ANS) are activated to facilitate neuroendocrine and cardiovascular adaptations to increase the survival of the organism. The HPA axis is a major system that mediates neuroendocrine responses to stress, resulting in the release of glucocorticoids (GCs) from the adrenal cortex (for review, see Herman et al., 2003). GCs are highly catabolic steroid hormones that affect multiple bodily functions, including energy mobilization, immune and reproductive functions, cognition, etc. The mature HPA axis exhibits a circadian rhythmicity with a peak around the time of waking and a trough during the quiescent time of the activity cycle. Alterations in the normal pattern of cortisol secretion (either higher or lower than normal) have been associated with both psychiatric and somatic illnesses, including depression, post-traumatic stress disorder (PTSD), fibromyalgia, hypertension, immunosupression, sexual dysfunction, among others (Chrousos and Gold, 1998; Heim et al., 2000a; McEwen, 1998; Yehuda et al., 2001b). Superimposed upon the diurnal pattern is activation of stressor-specific pathways that converge in the hypothalamus, where this information is integrated in the paraventricular nucleus (PVN) by parvocellular neurons expressing corticotropin releasing-factor (CRF). CRF is released from nerve endings in the median eminence in response to metabolic, psychological or physical threats and stimulates the release of adrenocorticotrophic hormone (ACTH) from the anterior pituitary. ACTH, in turn, stimulates the release of GCs (cortisol in primates, corticosterone in rats) from the adrenal cortex. These steroid hormones mobilize energy substrates during stress and regulate activity of the HPA axis via negative feedback at different levels via two types of corticosteroid receptors: (a) mineralocorticoid receptors (MR), with high affinity for GCs and important in proactive maintenance of HPA basal activity, and (b) glucocorticoid receptors (GR), with low affinity for GCs and primarily responsible for reactive negative feedback during the circadian peak and following an acute stressor.

In the rat brain, GR are broadly expressed, but particularly abundant in PVN and hippocampus, whereas MR is restricted primarily to hippocampus and septum. In primates, the distribution of MR and GR is roughly similar to that of the rat. For example, the high levels of both MR and GR in primate cortical areas, such as the prefrontal cortex, suggest a potentially important role of GCs in modulation of cognition, mood and social behavior, and indicate potential differences in the regulation of these output systems between rodents and primates (López et al., 1999; Patel et al., 2000; Sanchez et al., 2000). In addition to the different distributions in the adult brain, the postnatal ontogenetic patterns of neural MR and GR expression are also different in primates and in rats (Pryce et al., 2005a; Rosenfeld et al., 1988; Van Eekelen et al., 1991; Schmidt et al., 2003). These postnatal differences make sense if we consider that the primate brain is in a much more advanced state of development at birth than the rodent brain (Finlay and Darlington, 1995; Levitt, 2003), matching the higher maturational state of neonates in precocial species such as the macaque, in comparison to that of the altricial newborn rat. Understanding the postnatal changes in the expression of these two transcription factors – GR and MR – is important, because they regulate the expression of other genes during development, and because GR and MR's own expression (and, therefore, the activity of the HPA axis) is regulated by postnatal experiences, including quality of maternal care (Meaney and Szyf, 2005).

In rats and mice, levels of hippocampal GR expression are low at birth and increase markedly during development, reaching maximal levels in early adulthood (Suchecki et al., 1993a; Schmidt et al., 2003; Van Eekelen et al., 1991). Interestingly, early environment (including maternal care) can drastically affect GR expression, and the changes can be carried into adulthood, affecting HPA axis functionality, mainly its responsiveness to stress (Meaney and Szyf, 2005). However, in nonhuman primate brain, GR expression levels do not seem to change from birth to adulthood, although MR expression significantly peaks in the infant–juvenile period corresponding to weaning and more independent interaction with the social and physical environment (Pryce et al., 2005a). These authors propose that the timing of this increase in MR expression could make this gene a sensitive target for environmental effects

during weaning, affecting its later expression and, consequently, its regulation of basal HPA axis function. These findings could also explain the reports of long-term effects of early adversity on basal activity of the HPA axis in primates (discussed below), whereas in rodents the effects are more related to HPA axis reactivity to stress. Although this is a very interesting possibility, it has to be noted that marmosets and some other New World monkeys are characterized by high levels of steroid hormones, including cortisol (Coe et al., 1992), and these data may not resemble postnatal changes in Old World simian primates, apes or humans. Thus, further studies are necessary to understand the biological significance of these postnatal changes, as well as to demonstrate whether this is a species-specific phenomenon or it is actually common in other primate species.

In addition to its endocrine effects, CRF has a broad extrahypothalamic distribution in the CNS (Swanson et al., 1983), and its synthesis and function are affected by stress (Whitnall, 2001). Two CRF receptor subtypes (CRF1 and CRF2) have been identified, with different neuroanatomical distributions in primate and rat brain (Sanchez et al., 1999). These receptors mediate CRF effects on HPA function, behavioral and autonomic responses to stress, as well as on emotionality and cognitive function. Based on the distribution of corticosteroid and CRF receptors in the primate, sustained elevations of both GCs and CRF caused by exposure to chronic stress could impact neurobiological systems regulating not only HPA axis and autonomic function, but also mood, learning and memory processes, and different aspects of behavior.

The ontogeny of the HPA axis and its regulation has been reviewed in detail elsewhere (Walker et al., 2001), describing a wide variation among species in the development of the various components of the HPA axis and the onset of their functional activity. As described above for GR and MR, these developmental variations can partially explain different long-term effects of early adverse experiences on HPA axis function in rodents and primates. An important variation pointed out above is the different maturational state of the brain at birth, with rat pups exhibiting an equivalent brain development to human gestational age of 24 weeks (Plotsky et al., 2000). And, as described above for GR and MR, differences in postnatal development of specific components of the HPA system between rodents and primates could make "biological sense" because they match developmental milestones in each species. However, much research remains to be done in primates to get to the same sophisticated level of understanding of HPA axis postnatal development as in rodents. One aspect of postnatal development that is still unclear in primates is whether or not primate infants exhibit a stress hyporesponsive period (SHRP) equivalent to rats. Neonatal rats exhibit a reduced ability to mount an adult-like corticosterone response to stress from postnatal day (PND) 4–14, in comparison to adults (for review see Walker et al., 2001). This is important because the most robust effects of adverse caregiving models (e.g., maternal separation) are observed when imposed during this SHRP period. A hypothesis to explain these findings is that the HPA axis SHRP has the function of protecting the developing brain from high levels of stress hormones. A SHRP has also been described in other species including the rabbit, guinea pig and dog, with a shorter duration than in rats (days instead of one and a half week).

The presence of such a pituitary–adrenal refractory period in nonhuman primates and humans remains controversial. However, evidence is accumulating that in humans a functionally equivalent period may emerge over the first year and extend throughout childhood, maybe until puberty (for review see Gunnar and Vazquez, 2006; Tarullo and Gunnar, this volume). In rats, the HPA axis SHRP is mediated by appropriate levels of maternal care and stimulation, in particular by specific maternal behaviors such as feeding, licking and grooming (Rosenfeld et al., 1992; Suchecki et al., 1993b). If these aspects of maternal care are removed, the rat pup's HPA axis is not buffered anymore and can release high levels of ACTH and cortisol. Although in nonhuman primates, a similar SHRP has technically not been demonstrated, because infants exhibit adult-like basal and stress-induced levels of cortisol secretion (Bowman and Wolf, 1965; for reviews see Pryce et al., 2005b, Sanchez et al., 2001), maternal care does have an equivalent buffering effect on the infant's HPA axis response to stress. The specific aspects of maternal care that buffer the nonhuman primate infant's HPA axis responses are not known. But in rhesus macaques, we have some evidence, as in humans (Tarullo and Gunnar, this volume), that these aspects are complex and related to the sensitivity, responsiveness, active supervision, protectiveness, availability and support of the mother towards the infant, aspects previously demonstrated to lead to a secure infant attachment in macaques (Kondo-Ikemura and Waters, 1995). Thus, rhesus infants that normally exhibit significant elevations in ACTH and cortisol in response to stressful situations, do not have these physiological responses when their mother is present. But the key factor is not the mere presence of the mother, but the presence of a sensitive, responsive and protective mother, because the presence of a mother characterized by poor infant caregiving (e.g., high rejection, abusive, low protectiveness) does not buffer its infant's HPA response to stress (McCormack et al., 2003). Altogether, this evidence suggests that the development of the HPA axis in nonhuman primates is strongly regulated by social factors, particularly by aspects of maternal care, and that alterations in early caregiving can impact this neuroendocrine function via multiple mechanisms.

Adverse caregiving impacts HPA axis development

Maltreatment

Childhood abuse and neglect are very serious problems in our society. In addition to the obvious risk of physical injury or death, the emotional distress associated with acts of maltreatment negatively affects child development, leading to psychopathology and pathophysiology. For example, the experience of childhood maltreatment is associated with impaired social competence, emotional regulation and cognition, as well as an increased risk for the development of anxiety, mood and

addictive disorders (for reviews, see Glaser, 2000; Gunnar and Vazquez, 2006; Hart et al., 1995). Interestingly, this is not just a human problem. It has been reported in nonhuman primates as well. In populations of macaques and other monkeys, 2–10% of infants are physically abused by their mothers (Maestripieri, 1999; Maestripieri and Carroll, 1998b; Maestripieri et al., 1997a,b). This research has described infant abuse as manifested by violent behaviors by the mother such as dragging, hitting, crushing, throwing, stepping or sitting on the infant (Troisi and D'Amato, 1984; Maestripieri, 1998). These behaviors are aberrant and cause intense infant distress and sometimes serious injury or even death. Physical abuse is generally exhibited in the first 3 months postpartum, and repeated with successive infants. Infant abuse has higher prevalence in some matrilines and among related individuals, such as mothers, daughters or sisters, suggesting an intergenerational transmission along the maternal line. Whether this intergenerational transmission is due to experiential or learning factors, or to genetic predisposition is not clear yet, However, a recent study with individuals that were cross-fostered at birth, suggests that abuse gets perpetuated mainly as a result of the early experience (Maestripieri, 2005).

Thus, maternal maltreatment in rhesus constitutes a unique naturalistic animal model of infant maltreatment, currently used to define the behavioral, emotional and neuroendocrine outcomes of this early adverse experience in humans. To date, "this is the closest animal model of human child maltreatment" (Sanchez et al., 2001; Maestripieri, 1999). In these studies, animals are raised in a rich social and physical environment, with minimal outside disruption, which allows investigators to capture the effects of infant abuse on normal development and investigate the ability of abused animals to behaviorally and physiologically cope with novelty, threat and stress throughout their life span.

In humans, childhood maltreatment is a complex phenomenon with different manifestations (abuse: physical, sexual; neglect: physical, emotional), that often co-occur (Cicchetti, 1998; Glaser, 2000, 2002). In comparison to physical abuse, defining neglect in nonhuman primates has been more difficult. Like humans, nonhuman primate mothers demonstrate a broad spectrum of caregiving styles towards infants. And although some mothers show poor maternal care, it is difficult to determine whether that is harmful for the infant due to the lack of research on its developmental impact. Therefore, until recently, a conservative approach has been taken to define "neglect" in nonhuman primates as "complete abandonment of the infant by the mother" (Maestripieri and Carroll, 2000). Among rhesus and pigtail macaques, physical abuse rarely coexists with this definition of neglect, which tends to occur in 1.5% or 15.5% primiparous mothers, respectively (Maestripieri et al., 1997b; Maestripieri and Carroll, 1998b, 2000). However, recent research has reevaluated the definition of neglect in macaques to consider poor maternal care, low maternal responsiveness/sensitiveness and high levels of infant rejection exhibited by some macaque mothers as "neglectful" caregiving, and maybe comparable to neglect in humans (McCormack et al., in press). Interestingly, in addition to physical abuse, abusive macaque mothers often also display poor levels of maternal care towards their infants very early in life, including extremely high rates of infant rejection and contact breaks, lower than normal ventral contact and low protectiveness under threatening situations (Maestripieri, 1998; Maestripieri and Carroll, 1998a; McCormack et al., 2003, in press). The recent study by our group (McCormack et al., in press) used multidimensional scaling (MDS) to examine concurrently multiple aspects of maternal behavior and concluded that abusive mothers were also characterized by a highly "rejecting/anxious" maternal style, in clear opposition to the "nurturing/protective" maternal care exhibited by nonabusive mothers towards their infants. These deficits in maternal care occur very early in the infant's life (including the first month), when the rhesus macaque mother typically provides her infant with high levels of nurturance and protection (Hinde and Spencer-Booth, 1967; Suomi, 2005). Thus, physical abuse in macaques can co-occur with poor maternal care of the infant, characterized by high infant rejection, low responsiveness/sensitiveness to the infant and low protectiveness very early in life. This, too, is similar to human maltreatment where abuse frequently co-occurs with neglect (Cicchetti, 1998; Glaser, 2002).

The developmental impact of infant maltreatment in nonhuman primates is not fully understood. The information available so far suggests that physical abuse affects the normal socio-emotional development of the infant macaques, which grow up exhibiting behavioral signs of distress/irritability (high rates of tantrums and screams) and delayed social development (delayed independence from mom, as well as less exploration and play) (Maestripieri and Carroll, 1998a; Maestripieri et al., 2000, McCormack et al., in press). Interestingly, this behavioral profile is consistent with alterations previously reported in maltreated human children (Cicchetti, 1998; Crittenden and Ainsworth, 1997).

What do we know about the impact of infant maltreatment on neurobiological development of nonhuman primates and, in particular, on stress physiology and HPA axis function? Studies by our group are generating interesting findings that relate to human alterations. But, they could be better interpreted if we first review our current knowledge of the effects of early caregiving on the development of stress-related systems in other species.

In rodents, there is compelling evidence that the quality of maternal care (high versus low) dramatically impacts the development of neuroendocrine and neurobiological systems regulating stress physiology and adaptive behavior in the offspring. In particular, we know that low parental care or repeated maternal separations result in more anxious and stress-reactive offspring as adults (e.g., Meaney and Szyf, 2005; Sanchez et al., 2001). We know that the HPA axis of these adults is hyperresponsive to stress, and we also know now very sophisticated details about the molecular mechanisms mediating these effects, including how maternal care affects methylation levels of the GR gene and, therefore, its expression levels (Meaney and Szyf, 2005). Although in primates variation in parental care is also believed to result in long-term changes in the neuroendocrine and neurobiological development of the

offspring (e.g., Hinde, 1974), we have very limited experimental evidence of it, and we know even less about the mechanisms mediating the changes, or their developmental time course. There is evidence that extreme alterations in early caregiving (such as parental loss, maltreatment, or maternal depression) impact stress responses in human adulthood (Gunnar and Vazquez, 2006; Luecken and Lemery, 2004). And there is also some evidence that low responsive/sensitive parenting in humans is associated with larger cortisol responses to stress in toddlers (Gunnar et al., 1996) and enhanced fearfulness in infants, associated with more right frontal EEG asymmetry (Hane and Fox, in press). But much research is still needed to understand how early caregiving regulates the development of the HPA axis in primates.

Different studies have tried to identify (or hypothesize) the neurobiological mechanisms through which maltreatment produces its detrimental effects in humans (Gunnar and Vazquez, 2006; Tarullo and Gunnar, this volume). Many of these studies assume that childhood maltreatment is stressful/traumatic for the child and, therefore, activates main stress–response systems (HPA and autonomic nervous systems), with subsequent long-term, maladapatative changes for the developing organism. Although findings from adults with histories of childhood maltreatment are not consistent – partially explained by a high comorbidity with psychiatric disorders such as PTSD and depression, and the multifaceted nature of childhood maltreatment – low cortisol secretion throughout the circadian rhythm is commonly reported (Heim et al., 2000a; Roy, 2002; Yehuda et al., 2001b). Despite this basal hypofunctionality of the HPA axis, and the fact that the alterations at different levels of the HPA axis vary from study to study, the response to psychological stressors in adults with early adverse experiences can actually be sensitized (Heim et al., 2000b, 2001).

Data on children and adolescents with histories of childhood maltreatment are even more inconsistent and difficult to interpret (as reviewed by Tarullo and Gunnar in this volume). In addition to the ongoing maturation of neurobiological systems that regulate the stress responses, the confounding effects of comorbidity with psychiatric disorders and between different types of maltreatment, and the fundamental changes experienced during adolescence, some of these individuals are still living under adverse conditions. However, findings point more towards elevated than reduced basal cortisol levels in maltreated children, and hyperreactivity to challenges, particularly in those still living under stressful conditions (Bugental et al., 2003; De Bellis et al., 1999; Carrion et al., 2001, 2002; Hart et al., 1996; Kaufman, 1991, 1997).

Altogether, the human adult and children data suggest that in individuals that experience childhood maltreatment the pass of time and/or the transition through puberty and adolescence cause a switch in the alterations in HPA axis function. That is, their HPA axis seem to transition from an earlier phase characterized by hypercortisolism, likely due to the sustained exposure to adversity, to a later phase characterized by hypocortisolism—if adversity ceases (De Bellis et al., 1994; Yehuda et al., 2001a). Whether the later phase is a consequence of down-regulation of certain components of the HPA axis in response to continuous activation caused by chronic stress remains to be determined. So far, this is a still a working hypothesis, due to the lack of prospective, longitudinal studies in human populations that demonstrate what is really happening throughout development in individuals that experience childhood maltreatment.

However, findings from nonhuman primate models of early adverse experiences are starting to help put the pieces of this puzzle together. In fact, preliminary data by our research group from rhesus monkeys with histories of infant maltreatment support the view that early stress may produce initial elevations in cortisol, followed by lower than normal basal cortisol levels. Thus, abused infants exhibited higher basal levels of cortisol than controls during their first month of life (when maternal abuse was most frequent and intense), but lower levels thereafter (McCormack et al., 2003). At 6 months of age (equivalent to human toddler and preschool ages), pharmacological probes of the HPA axis demonstrated a blunted ACTH response to CRF in these maltreated infants, in comparison to controls. These findings reflect a down-regulation of CRF receptors in the pituitary, similarly to a previous report of HPA axis outcomes of negative/punitive parenting in another nonhuman primate, the common marmoset (*C. jacchus*) (Johnson et al., 1996), and strikingly similar to alterations detected in girls with history of childhood abuse (De Bellis et al., 1994). An explanatory hypothesis for all these findings is that a CRF overactivity due to emotional and/or physical stress in the early ages may have led to an adaptive down-regulation of CRF receptors in the anterior pituitary. The question of whether or not these alterations are carried into puberty and later into adulthood is still open, and constitutes an important part of our current longitudinal studies.

Other nonhuman primate models of early adverse caregiving

The imposition of unpredictable foraging demands in the mother is another paradigm used to disturb maternal caregiving in a different nonhuman primate: the bonnet macaque (*M. radiata*). This variable foraging demand paradigm (VFD: unpredictable alternation of periods of strenuous foraging with others in which free access to food is provided) produces profound effects on the mothers and their infants, suggesting that unpredictability represents an important factor in the outcomes (Andrews and Rosenblum, 1991). An immediate consequence of the stringent demand is a reduction in the amount of time that mothers respond to infant solicitations for contact and attention, which results in insecure patterns of attachment and delayed social development in the infants (Rosenblum and Andrews, 1994), similarly to some of the socioemotional alterations exhibited by abused rhesus macaque infants, and described above. As the infants mature, they become more fearful and hyperresponsive to stressful challenges (both behaviorally and physiologically). And these alterations are paralleled by persistent increases in cerebrospinal (CSF) levels of CRF (Coplan et al., 1996, 1998) and other endocrine alterations exhibited in anxiety and mood disorders (Coplan et al., 2000). So, in this model, we also find supporting

evidence that alterations in early adverse caregiving alter socioemotional development of the offspring, stress reactivity, and cause increased central CRF drive. Now, an important question in this and other models of early adverse caregiving is: how much of the socioemotional and central CRF alterations in the offspring are due to alterations in maternal care, and how much is due to the "stress" experienced by the infants during the different paradigms? This is a pressing question that researchers in this field can address and that can inform human studies.

Human and nonhuman primates exhibit similar responses to episodes of maternal separation, that is, intense behavioral and physiological responses including the activation of the HPA axis (Bowlby, 1968; Hinde and McGinnis, 1977). Since macaque infants are in close physical contact with their mother early in life, separation from the mother for even brief periods provokes robust behavioral and HPA axis responses, in addition to changes in specific neurochemical systems (Rilling et al., 2001; for review see Sanchez et al., 2001). Our group (Sanchez et al., 2005) has examined the short- and long-term consequences of repeated, brief mother–infant separations at 3–6 months of age as an early adverse caregiving protocol on rhesus monkey HPA axis function (including reactivity and basal diurnal rhythm), and emotional behavior (as measured by acoustic startle reactivity). This is part of a larger longitudinal study where the impact of this adverse rearing protocol is being examined throughout development.

First of all, the repeated, brief, unpredictable maternal separations produced significant short-term effects, such as alterations in mother–infant relationship, delayed infant social development and an initial sensitization of the infant's HPA axis to the repeated separations, manifested by higher cortisol responses at the end of the separation protocol, particularly in the female infants (Sanchez et al., 2005). The early adverse experience also produced persistent alterations in HPA axis function and emotional behavior (increased anxiety) of the offspring long after the last separation. In particular, the initial HPA axis sensitization to the repeated stress was followed by flattened diurnal rhythms of cortisol secretion in the juvenile period, caused by lower than normal morning cortisol levels, particularly in the females. Interestingly, the higher the cortisol response to the separations in the infants, the lower the basal cortisol levels as juveniles; that is, the "more flattened" the daytime rhythms of cortisol. These findings support the above hypothesis that repeated elevations in cortisol caused by early stress can result in lower rather than higher cortisol secretion later in life. They are consistent with similar findings of initial elevated cortisol levels and signs of sympathetic activation in infant marmosets subjected to repeated maternal separations (or "early deprivation") followed by lower than normal cortisol levels later in life (Dettling et al., 2002; Pryce et al., 2005b), and are comparable with disruptions of diurnal cortisol secretory rhythms reported in infant rhesus monkeys raised on cloth surrogate mothers (Boyce et al., 1995) and with the flattened daytime cortisol rhythms reported in children and adolescents with histories of early adverse caregiving, including maltreatment (Cicchetti and Rogosch, 2001a,b; Gunnar and Vazquez, 2001, 2006). Although the underlying mechanisms causing these flattened cortisol rhythms as well as their consequences are not clearly understood, there is emerging evidence that flat rhythms and low morning cortisol levels exert deleterious effects on general health (Sephton et al., 2000). Because cortisol levels affect metabolism, it has been proposed that the cortisol increase after awakening "provide the organism with sufficient energy to shift from a resting to an active state" (Pruessner et al., 1997), mobilizing energy substrates, increasing exploration and promoting acquisition and consolidation during learning processes (De Kloet, 1991; De Kloet et al., 1998).

The fact that females infants were more vulnerable to the adverse experience than males, at least in relation to the persistent HPA alterations associated with repeated stress experiences, is an interesting observation that could be related to clinical evidence of gender differences in the impact of early adversity on subsequent human HPA axis dysfunction and psychopathology (Heim and Nemeroff, 2001; Klimes-Dougan et al., 2001).

Now, how do the repeated separations affect HPA axis stress responsiveness? We are currently examining this question from different angles in rhesus monkeys. Previous studies have reported blunted behavioral and physiological responses to psychosocial stress (Dettling et al., 2002; Levine and Mody, 2003) and increased glucocorticoid feedback sensitivity (Lyons et al., 2000), although this pattern of alterations has not been detected in other studies (Capitanio et al., 1986). Therefore, the existing evidence is inconsistent and suggests the need for further studies to fully understand the long-term alterations in HPA axis stress-responsiveness in primates.

In summary, we have direct evidence that episodic disruption of mother–infant pairs by maternal separation was, indeed, a stressful experience for both the infant and the mother, and that infants did not habituate to the repeated separations. On the contrary, the infants HPA axis response to separation got sensitized with exposure to repeated bouts of separation from the mother, and the amplitude of the sensitization was associated with worse developmental outcomes in the juvenile period (i.e., more flattened cortisol daytime rhythms and higher levels of anxiety). However, it is important to note here that the quality of mother–infant relationship appeared to play an important role modulating the outcomes of the separation protocol. When mothers and infants were reunited after an episode of maternal separation, they reunited quickly and remained in close physical (usually ventral–ventral) contact for prolonged periods of time. This intensification of the mother–infant contact associated with stressful experiences is a well-described phenomenon in nonhuman primates (Coe et al., 1983; Hennessy, 1986; Hinde and McGinnis, 1977; Sanchez et al., 2001) and can be observed days, even months after the provocation ends. But the interesting finding in our study was that the strength of this mother–infant contact during the reunion seemed to "buffer" the infant's HPA axis reactivity to the separations, and was a predictor for better outcomes in HPA axis function later in life. This finding adds to the complexity of factors that need to be considered in these studies because they contribute to the individual variability in vulnerability to early adverse experiences: (a) infant's perceived stress, (b) maternal

buffering of the infant's stress response during reunion (i.e., social support), (c) sex, (d) genetic factors that increase or decrease vulnerability.

Conclusions

Research with nonhuman primate models of early adverse caregiving, including infant maltreatment, is crucial to fully understand the causes and consequences of similar experiences in humans. It is also essential to characterize the time course of biobehavioral alterations throughout development, using prospective, longitudinal studies performed under controlled experimental conditions and using invasive approaches that are unrealistic and unethical when studying human populations.

This review provides evidence that early disruptions in mother–infant relationship (e.g., infant maltreatment, maternal–infant separations) are stressful for nonhuman primate infants, and that the chronicity and "perceived" intensity of the stress experience are strong predictors of long-term alterations in stress physiology and socioemotional behavior. In the infant maltreatment and repeated maternal separation models, the early sustained HPA axis activations result in lower than normal functionality of the axis as the animals develop. However, there is also evidence that availability of a responsive and sensitive mother can buffer the deleterious effects of these adverse experiences.

In summary, the studies presented in this review support the hypothesis that early adverse caregiving is an important risk factor for the development of psychopathology and pathophysiology during childhood and adolescence. The goal of future studies is to examine the mechanisms underlying these alterations, the changes as the animals go through adolescence and into adulthood, as well as identify genetic factors that can increase individual vulnerability to the experience of early adversity.

Acknowledgments

This work was supported by NIH grants MH65046, MH58922 and RR-00165 base grant to the Yerkes National Primate Research Center. The author wishes to thank members of the "Early Experience, Stress and Prevention Science Network" (R21 MH65046) for stimulating discussions on this topic that influenced the views presented in this review.

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